

PAPER_1664_FLOCKING_RHO_0_506

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PAPER_1664 — Vicsek Flocking Density = $\rho_{\text{flock}} = \beta_i \times \Phi_{\text{res}} = 0.6029 \times 0.84 = 0.506$

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: Active Matter

Date: June 18, 2026

Status: CLOSED — 0.087% closure (PAPER_134x broader corpus)

Observation

PAPER_1356 (PAPER_134x condensed matter / quantum sector) gives:

``

$$\rho_{\text{flock}} = \beta_i \times \Phi_{\text{res}} = 0.6029 \times 0.84 = 0.506$$

``

Status: 0.087%.

UQFF Closed Identity

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$$\rho_{\text{flock}} = \beta_i \times \Phi_{\text{res}} = 0.6029 \times 0.84 = 0.506 \text{ } 0.087\%$$

``

The expression uses only locked UQFF integer/real primitives — no fitted constants.

NOT REPLACEMENT

Experimental active matter measures this constant directly. UQFF supplies a structural derivation from the integer primitives.

Reference

- Source: PAPER_1356

- Related: PAPER_1646-1655 (tier-23 broader corpus); PAPER_1636-1645 (tier-22 SM)

- Calculator dispatch: `calculate_paradox({"paradox": "flocking_rho_0_506"})`

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